



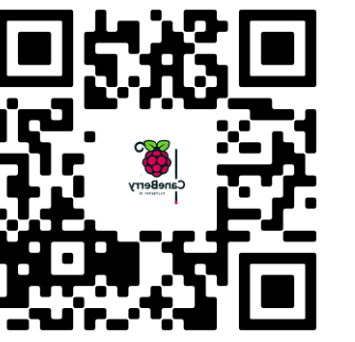
CANEBERRY

Walking without sight project

Members of the project

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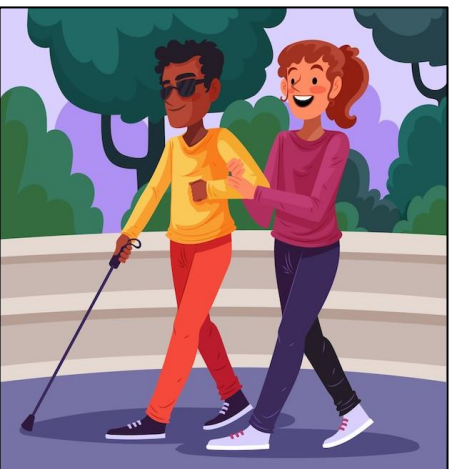


Why we should help VIP with new technology

Visually impaired people face many challenges in their daily life, for instance, walking in the street requires many special skills (like using the white cane, listening to the direction of the traffic). With new technologies, we can provide this more **efficient solution**, "CANE BERRY", that is easier to use.

Goals of the Caneberry tool

By using **accessible** components (cheap and easy to find everywhere) we help VIP to be **autonomous** : to walk straight and avoid obstacles on the ground.



« Pikisuperstar » - Freepik

How The Tool Works

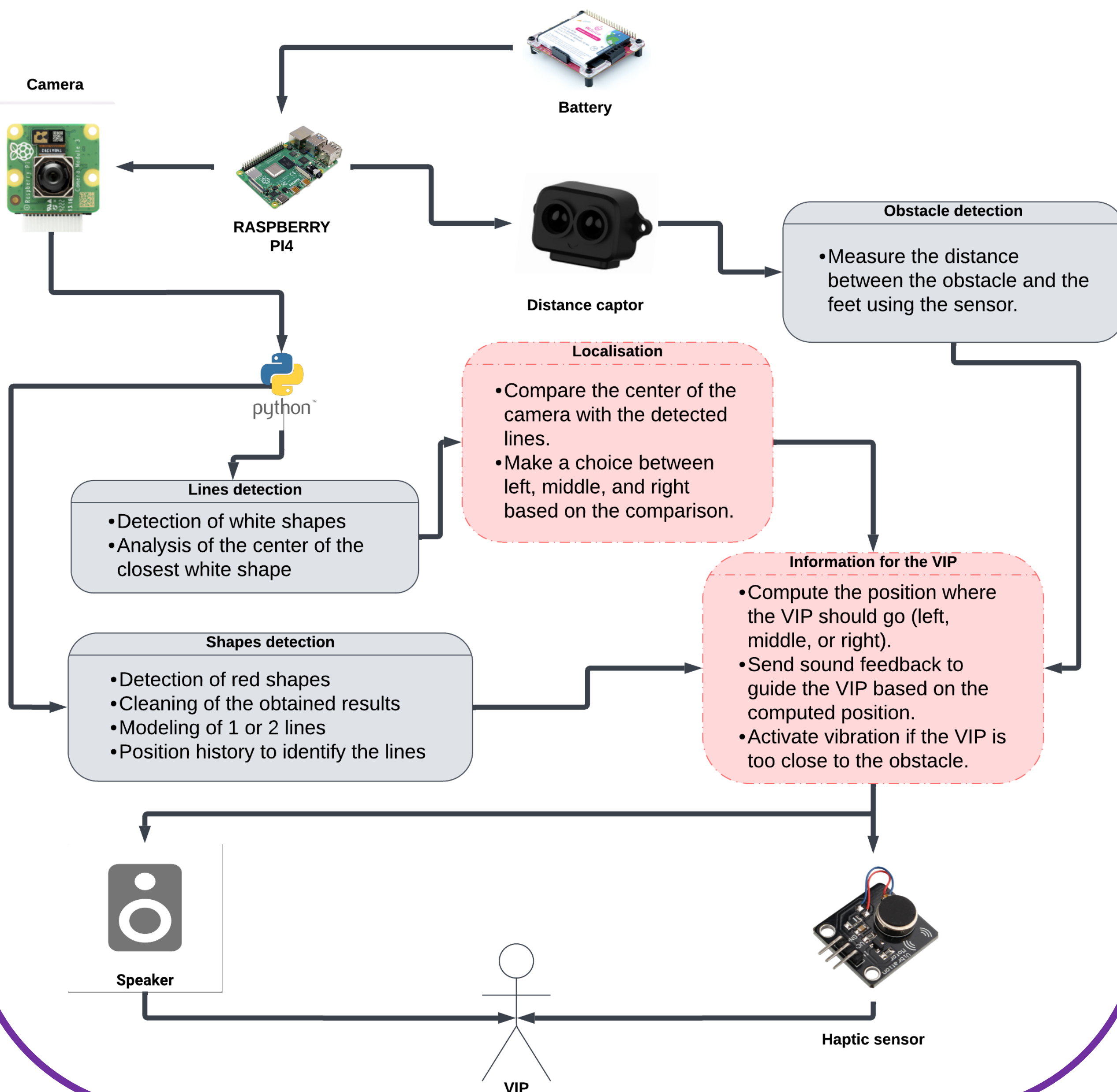
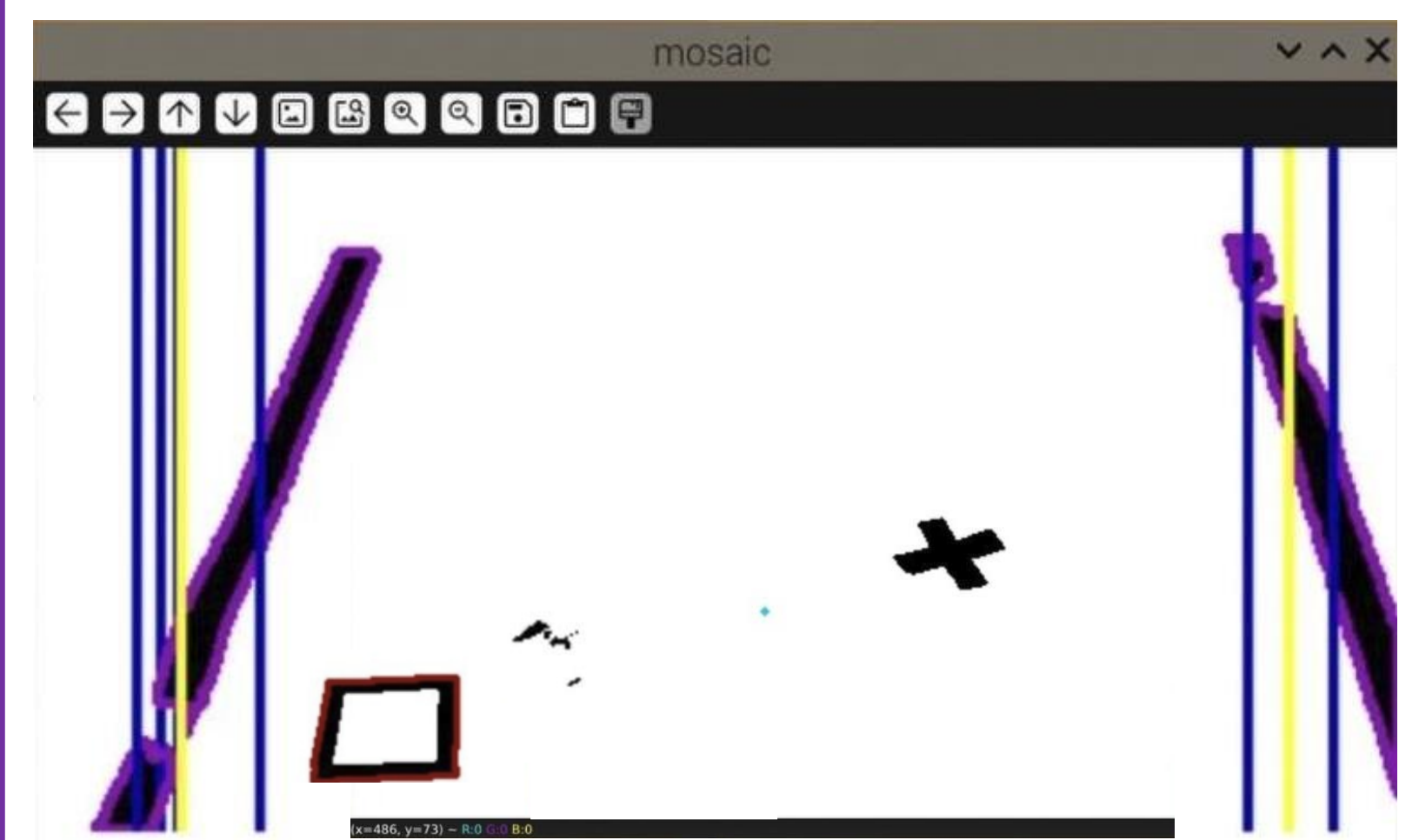


Illustration of lines and shapes detection



Lines and shapes detection

```

Shell
BETWEEN LINES
SQUARE, LEFT
BETWEEN LINES
SQUARE, LEFT
BETWEEN LINES
SQUARE, LEFT
BETWEEN LINES
SQUARE, LEFT
  
```

Information for the VIP

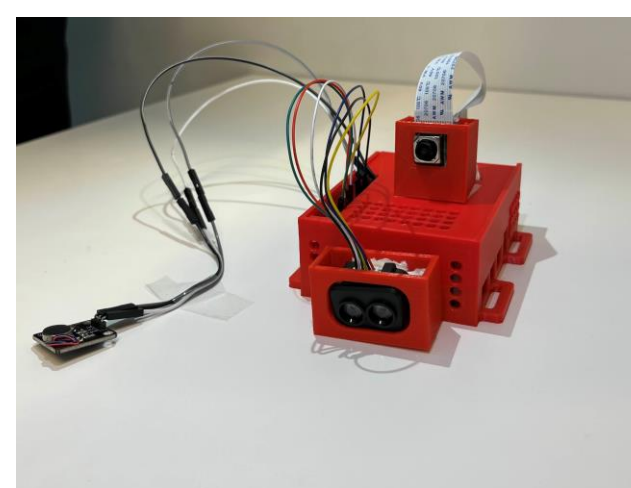
A few remarks:

- We analyze only the first shape since only the next shape is needed for **navigation**.
- We observe **potential issues** even after filtering, due to reflections.

Future Improvements

While the system performs well, there is still room for improvement:

- **Enhanced Accuracy:** Refining detection algorithms to improve precision for more complex scenarios with bad conditions
- **User Feedback:** Uses the feedback of tests to improve the guidance



Some results

Instances (nb of specific tests)	Rate of success with good conditions (25)	Rate of success with bad conditions (25)
50 - "cross detection"	90 %	75 %
50 - "square detection"	81 %	73 %
50- "barrier detection"	95 %	90 %
50 - "good position test"	72 %	68 %
50 - "whole tool test"	65 %	55 %

Table of rating of the tests with or without light reflection